



KUALA LUMPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

(formerly known as Infrastructure University Kuala Lumpur)

ASSIGNMENT 2: NETWORK ADMINISTRATION AND DEVICE SECURITY

Name:	Mohamed Mohamed Hussein
Student ID:	251925403
Course:	CSC1612 - Data Communication & Networking
University:	Kuala Lumpur University of Science and Technology
Lecturer:	VALERIANO DASALLA
Semester	Oct/Nov 2025
Date:	10 / 02 / 2026

1.0 Introduction

The goal of this assignment is to secure the network for the KL Branch. In Assignment 1, we built a network that worked, but it was not secure. A network without security is dangerous because unauthorized people can access it.

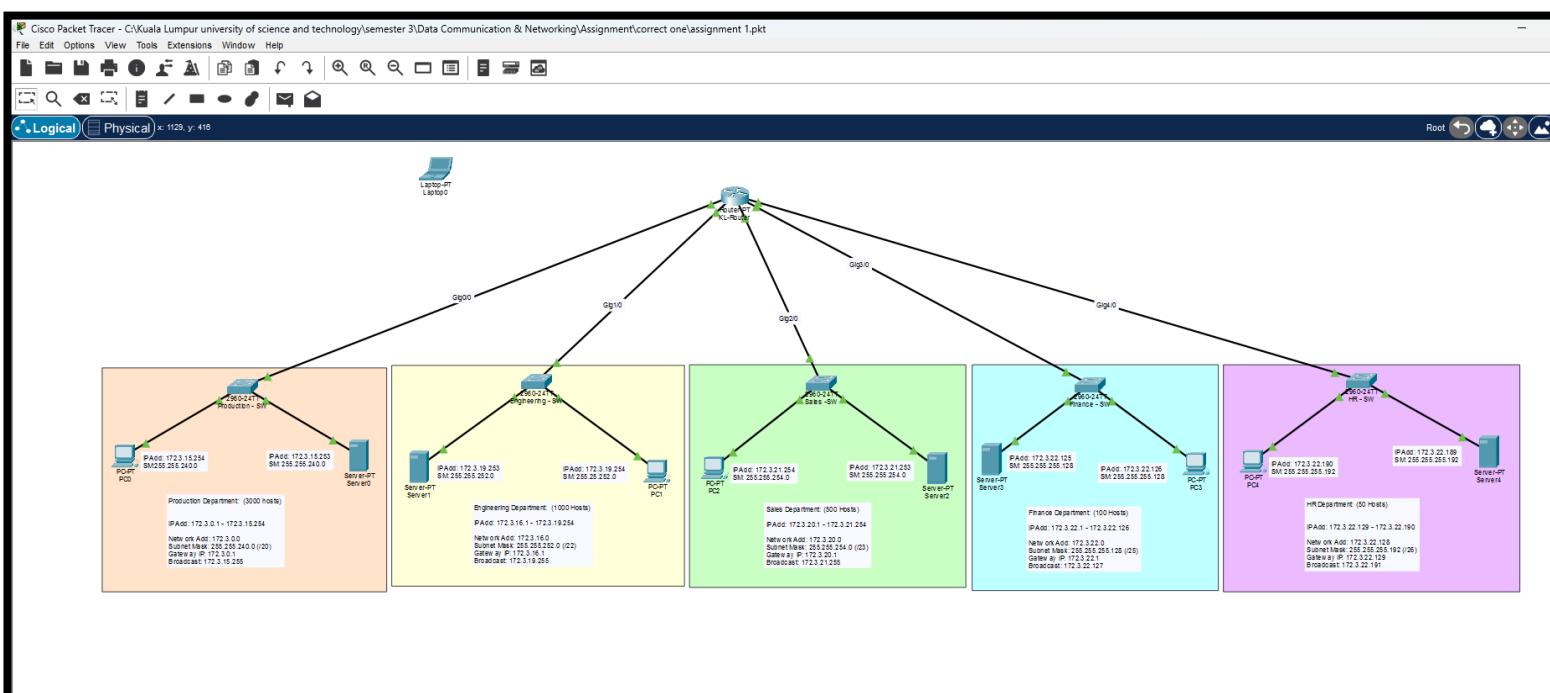
In this report, I will show how I secured the routers and switches. I followed the requirements to protect the devices from attacks. The main tasks were:

- Giving the devices correct names.
- Adding warning messages.
- Using passwords for all access points.
- Using SSH (Secure Shell) instead of Telnet for remote work.
- Turning off switch ports that are not in use.

This report documents every step I took to complete these tasks.

2.0 Network Topology

This is the final design of the network. It includes one router (KL-Router) and five switches for the different departments (Production, Engineering, Sales, Finance, and HR).



3.0 Basic Device Hardening

Hardening means making a system more secure. I started by configuring the basic settings on the KL-Router and the Department Switches.

3.1 Changing Hostnames

By default, the router is just named "Router". This is confusing for administrators. I changed the names to make them easy to identify.

- **Command used:** hostname KL-Router
- **Command used:** hostname Engineering-SW (and so on for others)

This helps us know exactly which device we are configuring.

```
!  
hostname KL-Router  
!  
!
```

```
KL-Router#  
KL-Router#  
KL-Router#enable|
```

3.2 Creating Warning Banners

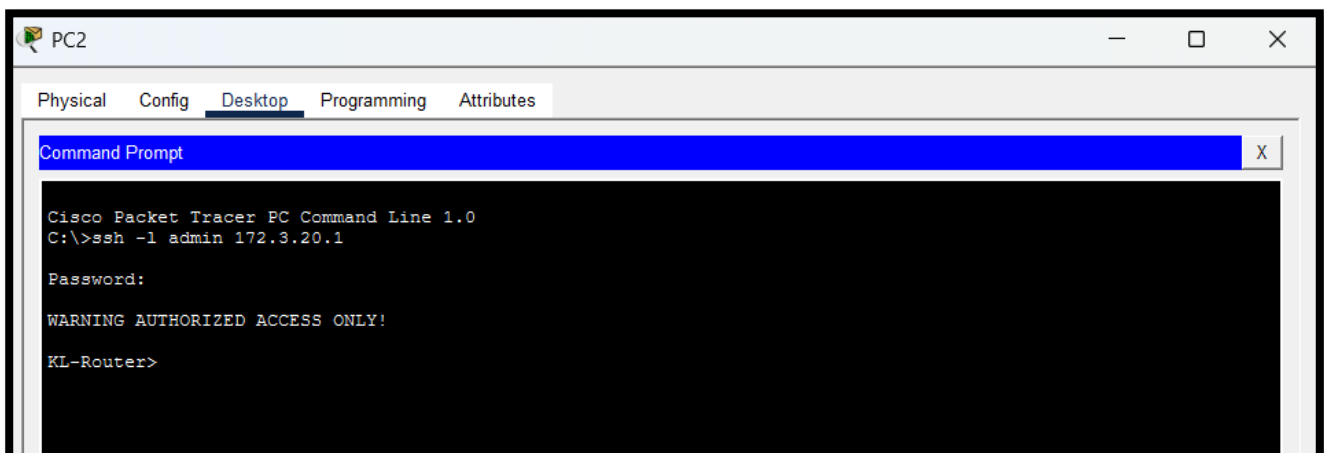
I configured a "Message of the Day" (MOTD). This is a banner that appears when someone tries to connect to the device.

The requirement was to warn unauthorized users.

- **Command used:** banner motd #WARNING AUTHORIZED ACCESS ONLY! #

This message is important for legal reasons. It tells attackers that they are not allowed to be here.

```
KL-Router>enable
KL-Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
KL-Router(config)#banner motd #WARNING AUTHORIZED ACCESS ONLY! #
KL-Router(config)#
KL-Router(config)#
KL-Router(config)#enable secret klust123
KL-Router(config)#line console 0
KL-Router(config-line)#password klust
KL-Router(config-line)#login
KL-Router(config-line)#exit
KL-Router(config)#
KL-Router(config)#service password-encryption
KL-Router(config)#
KL-Router(config)#end
KL-Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
KL-Router#
```



3.3 Encrypting Passwords

When we set passwords, they are normally saved as plain text. This is risky because anyone who sees the configuration file can read the passwords.

To fix this, I used a special command that encrypts all passwords in the file.

- **Command used:** service password-encryption

Now, if someone looks at the configuration, they will only see random numbers instead of the real password.

4.0 Securing Access (Management Plane)

The "Management Plane" is how we control the device. We need to make sure only the administrator can access it.

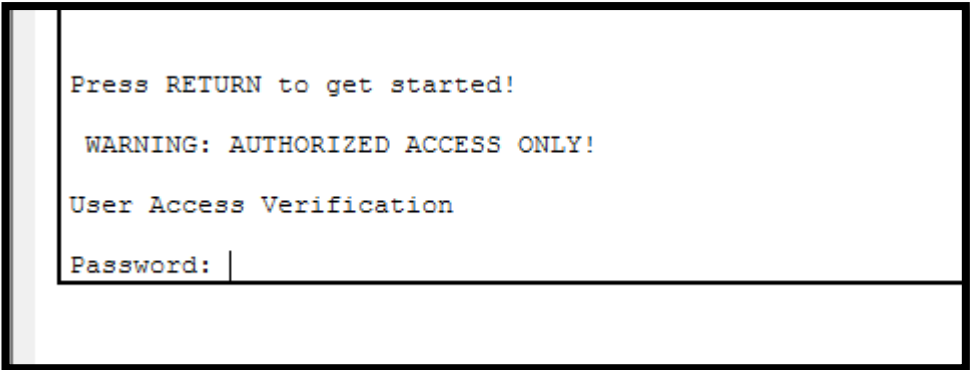
4.1 Securing the Console Port

The console port is the physical plug on the back of the router. If I do not secure it, anyone in the server room can plug in a cable and take control.

I added a password to the console line.

- **Command:** line console 0
- **Command:** password klust
- **Command:** login

Now, the router asks for a password immediately when connected.



```
Press RETURN to get started!  
  
WARNING: AUTHORIZED ACCESS ONLY!  
  
User Access Verification  
Password: |
```

4.2 Securing Privileged EXEC Mode

After logging in, a user is in "User Mode" (>). To make changes, they must go to "Privileged Mode" (#). This is the most important part of the router to protect.

I used the enable secret command. This is better than enable password because secret uses MD5 encryption, which is much stronger.

- **Command:** enable secret klust123

```
KL-Router>enable
KL-Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
KL-Router(config)#banner motd #WARNING AUTHORIZED ACCESS ONLY! #
KL-Router(config)#
KL-Router(config)#
KL-Router(config)#enable secret klust123
KL-Router(config)#line console 0
KL-Router(config-line)#password klust
KL-Router(config-line)#login
KL-Router(config-line)#exit
KL-Router(config)#
KL-Router(config)#service password-encryption
KL-Router(config)#
KL-Router(config)#end
KL-Router#
%SYS-5-CONFIG_I: Configured from console by console
wr
Building configuration...
[OK]
KL-Router#
```

5.0 Remote Management (SSH Configuration)

We need to manage the router from a distance (remotely). The assignment requires using SSH instead of Telnet.

Why SSH?

Telnet sends data in plain text. If a hacker intercepts the connection, they can read our passwords. SSH (Secure Shell) encrypts the connection, so the data is safe.

5.1 Creating a Domain and User

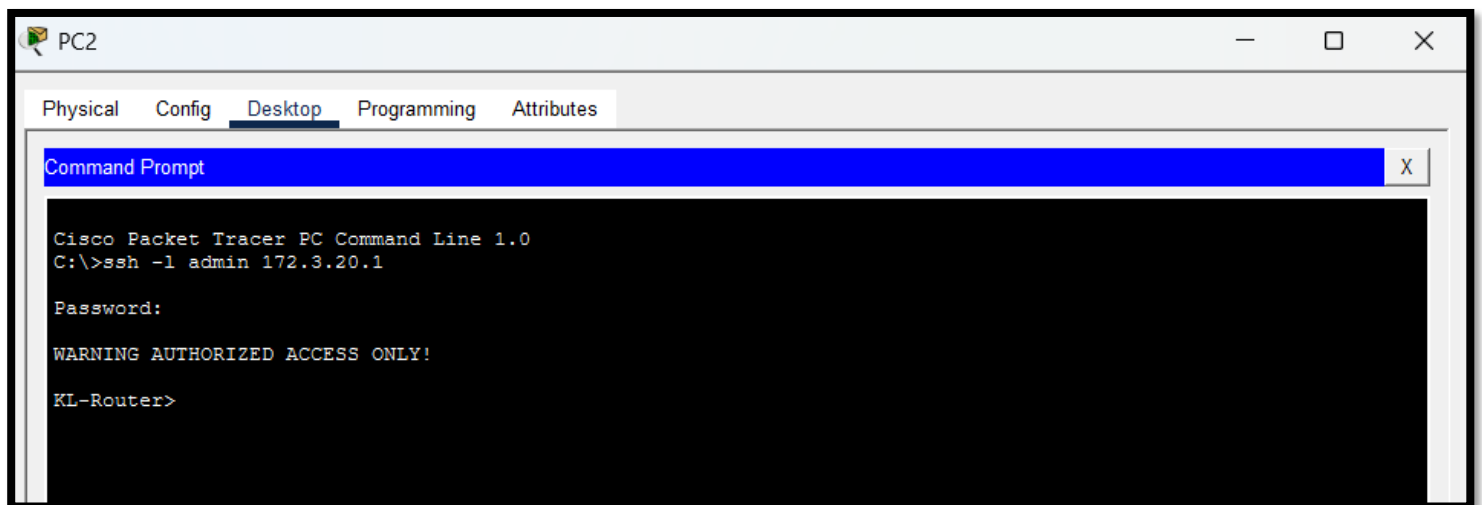
SSH requires a domain name and crypto keys to work.

1. I set the domain: ip domain-name company.com
2. I generated the security keys: crypto key generate rsa (1024 bits).
3. I created a username and password for the admin:
 - **Username:** admin
 - **Password:** klust

5.2 Enabling SSH on VTY Lines

Finally, I configured the virtual lines (VTY) to only accept SSH connections. This blocks Telnet completely.

- **Command:** transport input ssh



6.0 Port Security (Switch Hardening)

This section is about the switches. We need to make sure people cannot just plug any computer into the wall sockets.

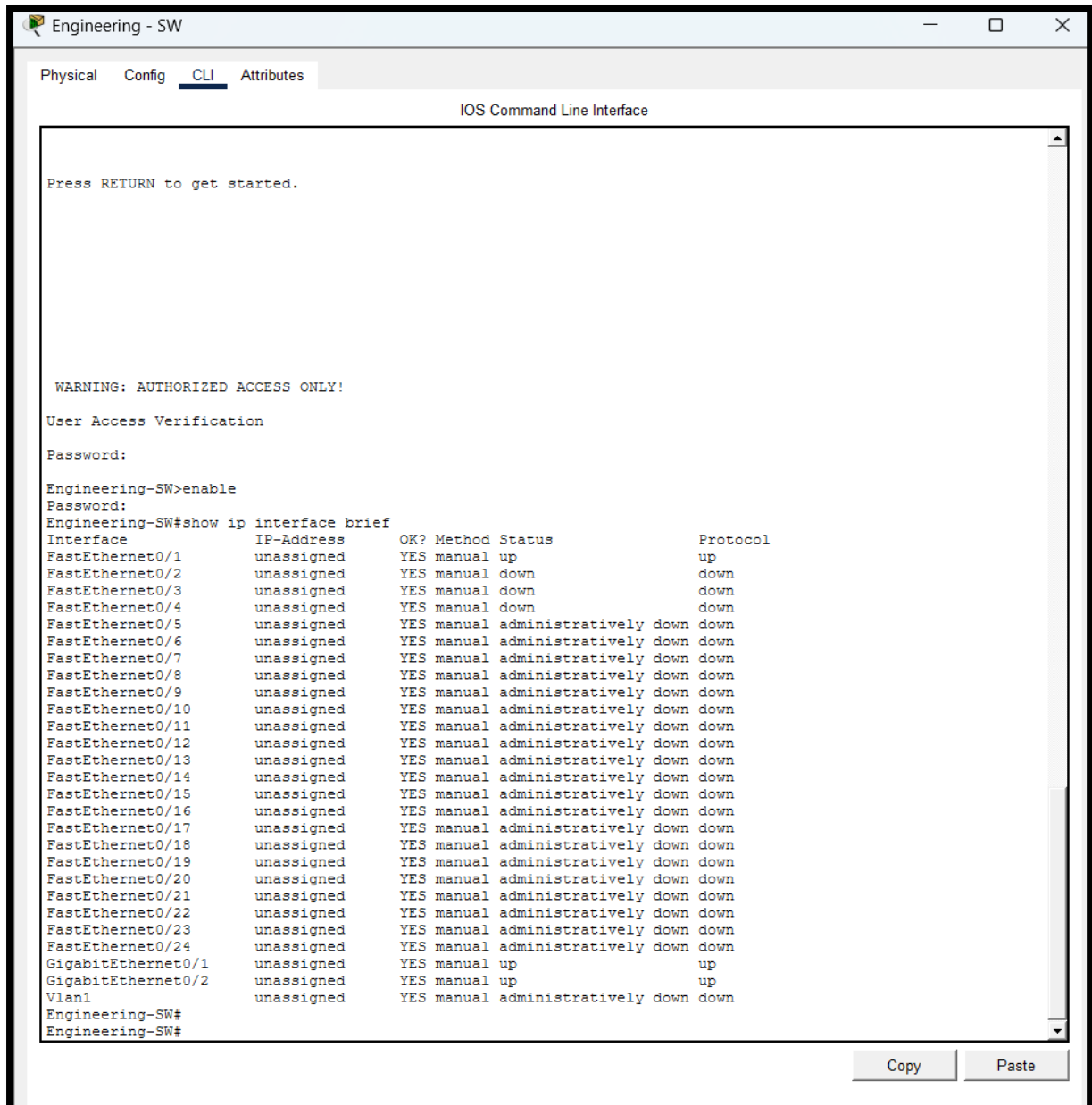
6.1 Shutting Down Unused Ports

On the Engineering Switch, we only use a few ports. However, the switch has 24 ports. If ports 5-24 are left "up," anyone can connect to them.

I manually shut down all the empty ports to prevent "plug-and-play" access.

- **Command:** interface range fa0/5 - 24
- **Command:** shutdown

I verified this using the show ip interface brief command. As you can see in the screenshot below, the unused ports say "administratively down."



```
Engineering - SW
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started.

WARNING: AUTHORIZED ACCESS ONLY!
User Access Verification
Password:
Engineering-SW#enable
Password:
Engineering-SW#show ip interface brief
Interface      IP-Address      OK? Method Status                Protocol
FastEthernet0/1 unassigned      YES manual up                    up
FastEthernet0/2 unassigned      YES manual down              down
FastEthernet0/3 unassigned      YES manual down              down
FastEthernet0/4 unassigned      YES manual down              down
FastEthernet0/5 unassigned      YES manual administratively down down
FastEthernet0/6 unassigned      YES manual administratively down down
FastEthernet0/7 unassigned      YES manual administratively down down
FastEthernet0/8 unassigned      YES manual administratively down down
FastEthernet0/9 unassigned      YES manual administratively down down
FastEthernet0/10 unassigned      YES manual administratively down down
FastEthernet0/11 unassigned      YES manual administratively down down
FastEthernet0/12 unassigned      YES manual administratively down down
FastEthernet0/13 unassigned      YES manual administratively down down
FastEthernet0/14 unassigned      YES manual administratively down down
FastEthernet0/15 unassigned      YES manual administratively down down
FastEthernet0/16 unassigned      YES manual administratively down down
FastEthernet0/17 unassigned      YES manual administratively down down
FastEthernet0/18 unassigned      YES manual administratively down down
FastEthernet0/19 unassigned      YES manual administratively down down
FastEthernet0/20 unassigned      YES manual administratively down down
FastEthernet0/21 unassigned      YES manual administratively down down
FastEthernet0/22 unassigned      YES manual administratively down down
FastEthernet0/23 unassigned      YES manual administratively down down
FastEthernet0/24 unassigned      YES manual administratively down down
GigabitEthernet0/1 unassigned      YES manual up                    up
GigabitEthernet0/2 unassigned      YES manual up                    up
Vlan1          unassigned      YES manual administratively down down
Engineering-SW#
Engineering-SW#
```

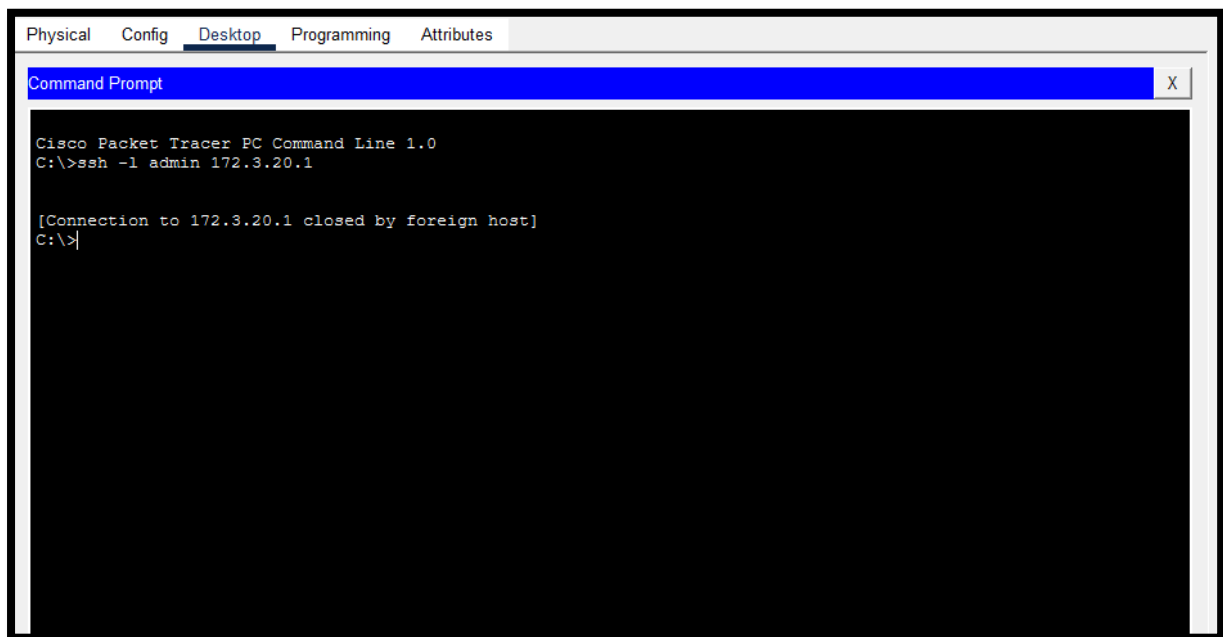
7.0 Verification and Testing

To prove the security works, I performed a test from PC2 in the Sales Department.

1. I opened the Command Prompt on PC2.
2. I typed: `ssh -l admin 172.3.20.1`
3. The router asked for my password.
4. After entering the password, I saw the banner: **"WARNING AUTHORIZED ACCESS ONLY!"**

This proves that:

- Remote access is working.
- The connection is using SSH (Secure).
- The banner is correctly displayed.



The screenshot shows a Cisco Packet Tracer PC Command Line window. The window has a title bar with tabs: Physical, Config, Desktop, Programming, and Attributes. The 'Desktop' tab is selected. The window title is 'Command Prompt'. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 172.3.20.1

[Connection to 172.3.20.1 closed by foreign host]
C:\>
```

8.0 Conclusion

In this assignment, I successfully secured the network infrastructure for the KL Branch.

I learned that basic configuration is not enough; we must also "harden" the devices. By using encrypted passwords, warning banners, and SSH, the management plane is now safe. Also, by shutting down unused ports on the switches, I have reduced the risk of physical attacks. The network is now ready for use.